NE723 Project 2: Quasi-Diffusion Acceleration for S_N in 1D Slab Geometry

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**** Source code available at the GitHub link below. ****

`https://github.com/zbevans3/NE723/tree/main/project-2`

1 Introduction

In this work, a one-speed discrete ordinates (S_N) neutron transport solver for steady-state, one-dimensional slab geometry with isotropic scattering and multiplying medium was developed that can be applied for fixed source and eigenvalue problems. Fixed source problems are described by Equation 1, which includes fission and an external source term.

$$\mu\frac{\partial\psi(x,\mu)}{\partial x}+\Sigma_t(x,\mu)\,\psi(x,\mu)=\frac{1}{2}\big(\Sigma_s(x)+\nu_f(x)\Sigma_f(x)\big)\int_{-1}^{1}\psi(x,\mu')d\mu'+\frac{1}{2}q(x) \tag{1}$$

where

$$x\in[0,X],\quad \mu\in[-1,1]$$

$$\psi(0,\mu)=\begin{cases}\psi_{in}^{+}(\mu) & \text{for the boundary with incoming particles}\\ \psi(0,-\mu) & \text{for the reflecting boundary}\end{cases}\quad \text{for } \mu>0$$

$$\psi(X,\mu)=\psi_{in}^{-}(\mu)\quad \text{for } \mu<0$$

$$\mu\frac{\partial\psi(x,\mu)}{\partial x}+\Sigma_t(x,\mu)\,\psi(x,\mu)=\frac{1}{2}\big(\Sigma_s(x)+\nu_f(x)\Sigma_f(x)\big)\int_{-1}^{1}\psi(x,\mu')d\mu' \tag{2}$$

where

$$x\in[0,X],\quad \mu\in[-1,1]$$

$$\psi(0,\mu)=\psi_{in}^{+}(\mu)=0\quad \text{for}\quad \mu>0$$

$$\psi(X,\mu)=\psi_{in}^{-}(\mu)=0\quad \text{for}\quad \mu<0$$

The source-free transport equation, Equation 2, describes the calculation solved in this work. The eigenvalue calculation can use reflective or vacuum boundary conditions, but only vacuum are necessary. The transport equations are discretized using Gauss–Legendre angular quadrature and the Linear Discontinuous Galerkin (LD) or Lumped Linear Discontinuous (LLD) spatial methods. The Source Iteration method and a power iteration are used to solve fixed source and eigenvalue problems, respectively. For Project 2, the transport solver was improved through implementing Quasi-Diffusion (QD) acceleration to speed up source convergence. The effectiveness of QD was investigated through four test problems with varying levels of diffusivity, which is a driving factor for slow source convergence in unaccelerated transport.

2 Method

2.1 Discrete Ordinates Method

The Discrete Ordinates Method (S_N) is a determinisitc method of solving the integro-differential transport equation by dividing the angular domain into discrete angles that have a corresponding weight. The flux is then numerically integrated by multiplying an ordinate's angular flux and the quadrature weight. Applying the Discrete Ordinates Method to Equation 1, the resulting equation in 1D slab geometry in steady-state, with isotropic becomes:

$$\mu_m \frac{d\psi_m(x)}{dx} + \Sigma_t(x)\psi_m(x) = \frac{1}{2}\Big(\Sigma_s(x) + \nu_f(x)\Sigma_f(x)\Big) \sum_{m'=1}^{M} \psi_{m'}(x)\, w_{m'} + \frac{1}{2}\, q(x) \tag{3}$$

$$\psi_m(0) = \psi_{in}^{+}(\mu_m) \quad \text{for } \mu_m > 0 \quad , \quad \psi_m(X) = \psi_{in}^{-}(\mu_m) \quad \text{for } \mu_m < 0$$

where

$$\psi_m(x) = \psi(x, \mu_m),$$

$$\phi(x) = \sum_{m=1}^{M} \psi(x, \mu_m)\, w_m,$$

$$J(x) = \sum_{m=1}^{M} \mu_m\, \psi(x, \mu_m)\, w_m,$$

The choice of angular quadrature (μ_m, w_m) is problem dependent, but in general, increasing quadrature order leads to a better approximation of the neutron flux. Equation 1 is discrete in angle but is still continuous in space. For finite computation (i.e. computers), the domain must be discretized into a finite number of cells. Several methods exist for discretizing the S_N equations in space, including the Linear Discontinuous Galerkin (LD) method and the Lumped Linear Discontinuous (LLD) method. LD and LLD both consist of zeroth and first spatial moment equations and follow the same derivation. In the moment form this changes a single coefficient, α, in the first moment equation, where $\alpha = 1$ for LD and $\alpha = 3$ for LLD. The general equations applied to Equation 3 for LD and LLD are given as:

$$\mu_m\big(\psi_{m,i+1/2} - \psi_{m,i-1/2}\big) + \Sigma_{t,i}\, \Delta x_i\, \psi_{m,i} = \frac{\Delta x_i}{2}\Big(\big(\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i}\big)\phi_i + q_i\Big) \tag{4}$$

$$3\mu_m\big(\psi_{m,i+1/2} + \psi_{m,i-1/2} - 2\psi_{m,i}\big) + \alpha\, \Sigma_{t,i}\, \Delta x_i\, \widehat{\psi}_{m,i} = \alpha\, \frac{\Delta x_i}{2}\Big(\big(\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i}\big)\widehat{\phi}_i + \widehat{q}_i\Big) \tag{5}$$

$$\alpha = \begin{cases} 1 & , \quad \text{LD}, \\ 3 & , \quad \text{LLD}, \end{cases}$$

with the upwinding condition:

$$\psi_{m,i\mp1/2} = \begin{cases} \psi_{m,i} - \widehat{\psi}_{m,i} & , \quad \text{for } \mu_m < 0, \\ \psi_{m,i} + \widehat{\psi}_{m,i} & , \quad \text{for } \mu_m > 0, \end{cases}$$

$$i = 1, ..., N_{cells}$$

where the cell average scalar flux is:

$$\phi_i = \sum_{m=1}^{M} \psi_{m,i}\, w_m,$$

and the linear spatial moment of the scalar flux is:

$$\widehat{\phi}_i = \sum_{m=1}^{M} \widehat{\psi}_{m,i}\, w_m.$$

Cell index $i \pm 1/2$ implies cell edge quantities, while i is the cell average. Solving the angular flux in a cell or cell edge requires sweeping through the mesh, starting from a known boundary condition. When stepping through the mesh, the incoming flux for cell i in direction μ_m is known, therefore the four unknowns in Equation 4 and Equation 5 are reduced to three. The upwinding condition can then be used to recast the two equations into a system of two unknowns, $\psi_{m,i}$ and $\widehat{\psi}_{m,i}$.

For the positive direction ($\mu_m > 0$), the sweep begins at $x = 0$ and proceeds with $\psi_{m,i-1/2}$ being known. After substituting the upwind condition $\psi_{m,i+1/2} = \psi_{m,i} + \widehat{\psi}_{m,i}$ and dividing by μ_m, Equation 4 and Equation 5 become:

$$\psi_{m,i} + \widehat{\psi}_{m,i} + \frac{\Sigma_{t,i}\,\Delta x_i}{\mu_m}\,\psi_{m,i} = \frac{1}{\mu_m}\,Q_i + \psi_{m,i-1/2} \tag{6}$$

$$3\psi_{m,i} + 3\widehat{\psi}_{m,i} - 6\psi_{m,i} + \alpha\,\frac{\Sigma_{t,i}\,\Delta x_i}{\mu_m}\,\widehat{\psi}_{m,i} = \frac{\alpha}{\mu_m}\,\widehat{Q}_i - 3\psi_{m,i-1/2} \tag{7}$$

where

$$Q_i = \frac{\Delta x_i}{2}\left(\left(\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i}\right)\phi_i + q_i\right)$$

$$\widehat{Q}_i = \frac{\Delta x_i}{2}\left(\left(\Sigma_{s,i} + \nu_{f,i}\Sigma_{f,i}\right)\widehat{\phi}_i + \widehat{q}_i\right)$$

After combining like terms, a linear system can be formed to find $\psi_{m,i}$ and $\widehat{\psi}_{m,i}$:

$$\begin{bmatrix} 1+\beta_{m,i} & 1 \\ -3 & 3+\alpha\beta_{m,i} \end{bmatrix} \begin{bmatrix} \psi_{m,i} \\ \widehat{\psi}_{m,i} \end{bmatrix} = \begin{bmatrix} \frac{1}{\mu_m}Q_i + \psi_{m,i-1/2} \\ \frac{\alpha}{\mu_m}\widehat{Q}_i - 3\psi_{m,i-1/2} \end{bmatrix} \tag{8}$$

where

$$\beta_{m,i} = \frac{\Sigma_{t,i}\Delta x_i}{\mu_m}$$

Inverting the matrix system leads to equations for the positive μ_m direction that need to be solved for cell i to obtain the cell-wise quantities. $\psi_{m,i+1/2}$ can then be computed by the upwind condition

from the newly obtained quantities.

$$\boxed{\begin{aligned}
\psi_{m,i} &= \frac{(3+\alpha\beta_{m,i})Q_i - \alpha\widehat{Q}_i}{\mu_m\, G_{m,i}} + \frac{6+\alpha\beta_{m,i}}{G_{m,i}}\psi_{m,i-1/2} \\
\widehat{\psi}_{m,i} &= \frac{3Q_i + \alpha(1+\beta_{m,i})\widehat{Q}_i}{\mu_m\, G_{m,i}} - \frac{3\beta_{m,i}}{G_{m,i}}\psi_{m,i-1/2} \\
G_{m,i} &= \alpha\beta_{m,i}^2 + (3+\alpha)\beta_{m,i} + 6 \\
\psi_{m,i+1/2} &= \psi_{m,i} + \widehat{\psi}_{m,i}
\end{aligned}} \tag{9}$$

For the negative direction ($\mu_m < 0$), the sweep begins at $x = X$ with $\psi_{m,i+1/2}$ being known. The process of finding the linear system for the positive direction is then repeated for the negative direction, except that the upwind condition is $\psi_{m,i-1/2} = \psi_{m,i} - \widehat{\psi}_{m,i}$. Upon substitution and dividing by μ_m, the set of equations becomes:

$$-\psi_{m,i} + \widehat{\psi}_{m,i} + \frac{\Sigma_{t,i}\,\Delta x_i}{\mu_m}\,\psi_{m,i} = \frac{1}{\mu_m}\,Q_i - \psi_{m,i+1/2} \tag{10}$$

$$3\psi_{m,i} - 3\widehat{\psi}_{m,i} - 6\psi_{m,i} + \alpha\,\frac{\Sigma_{t,i}\,\Delta x_i}{\mu_m}\,\widehat{\psi}_{m,i} = \frac{\alpha}{\mu_m}\,\widehat{Q}_i - 3\psi_{m,i+1/2} \tag{11}$$

which results in the linear system:

$$\begin{bmatrix} \beta_{m,i} - 1 & 1 \\ -3 & \alpha\beta_{m,i} - 3 \end{bmatrix} \begin{bmatrix} \psi_{m,i} \\ \widehat{\psi}_{m,i} \end{bmatrix} = \begin{bmatrix} \frac{1}{\mu_m}Q_i - \psi_{m,i+1/2} \\ \frac{\alpha}{\mu_m}\widehat{Q}_i - 3\psi_{m,i+1/2} \end{bmatrix} \tag{12}$$

Thus, the equations for each of the unknowns for cell i when sweeping in the negative direction ($\mu_m < 0$) are:

$$\boxed{\begin{aligned}
\psi_{m,i} &= \frac{(\alpha\beta_{m,i} - 3)Q_i - \alpha\widehat{Q}_i}{\mu_m\, G_{m,i}} + \frac{6-\alpha\beta_{m,i}}{G_{m,i}}\psi_{m,i+1/2} \\
\widehat{\psi}_{m,i} &= \frac{3Q_i + \alpha(\beta_{m,i}-1)\widehat{Q}_i}{\mu_m\, G_{m,i}} - \frac{3\beta_{m,i}}{G_{m,i}}\psi_{m,i+1/2} \\
G_{m,i} &= \alpha\beta_{m,i}^2 - (3+\alpha)\beta_{m,i} + 6 \\
\psi_{m,i-1/2} &= \psi_{m,i} - \widehat{\psi}_{m,i}
\end{aligned}} \tag{13}$$

Note that $\beta_{m,i} < 0$ for $\mu_m < 0$. For $\alpha = 1$, Equation 9 and Equation 13 reduce to the standard LD sweep equations.

2.2 Quasi-Diffusion Acceleration

The Quasi-Diffusion method is a nonlinear acceleration scheme that solves a simpler set of equations (low order, LO) after the completion of a transport solve (low order, HO) in order to obtain a better approximation of the scalar flux and subsequently, a better scattering distribution. The method includes exact closures that are derived and obtained by the transport solution to ensure consistency between the two equations. The 1D steady-state, one speed, low order QD equations in continuous space are given as:

$$\frac{d}{dx}J(x) + \big(\Sigma_t - (x)\Sigma_s(x)\big)\phi(x) = q(x) \tag{14}$$

$$\frac{d}{dx}\bigg(E(x)\,\phi(x)\bigg) + \Sigma_t(x)J(x) = 0 \tag{15}$$

with boundary conditions:

$$J(0) = C_L(\phi(0) - \phi^+_{in}) + J^+_{in} \tag{16}$$

$$J(X) = C_R(\phi(X) - \phi^-_{in}) + J^-_{in} \tag{17}$$

The Eddington factor ($E(x)$) and boundary factors (C_L, C_R) are the closure that relates the HO to LO solution.

$$E(x) = \frac{\int_{-1}^{1}\mu^2\psi(x,\mu)d\mu}{\int_{-1}^{1}\psi(x,\mu)d\mu} \tag{18}$$

$$C_L = \frac{\int_{-1}^{0}\mu\psi(0,\mu)d\mu}{\int_{-1}^{0}\psi(0,\mu)d\mu} \tag{19}$$

$$C_R = \frac{\int_{0}^{1}\mu\psi(X,\mu)d\mu}{\int_{0}^{1}\psi(X,\mu)d\mu} \tag{20}$$

For the low-order QD equations (LOQD), they are independently discretized using a second-order finite volume scheme, and the LOQD solution is signified by the LO superscript. The Eddington and boundary factors (E, C_L, C_R) are computed from the high-order solution and carry no superscript. Starting with the balance equation, integrating over the i-th spatial cell, where $\Delta x_i = x_{i+1/2} - x_{i-1/2}$, leads to:

$$J^{LO}_{i+1/2} - J^{LO}_{i-1/2} + (\Sigma_{t,i} - \Sigma_{s,i})\,\Delta x_i\,\phi^{LO}_i = q_i\,\Delta x_i \tag{21}$$

The first moment equations are integrated over each half of a cell, where $x_i = 0.5(x_{i+1/2} + x_{i-1/2})$. For the left half of cell i:

$$E_i\,\phi_i^{LO} - E_{i-1/2}\,\phi_{i-1/2}^{LO} + \frac{1}{2}\Sigma_{t,i}\,\Delta x_i\,J_{i-1/2}^{LO} = 0 \tag{22}$$

Then for the right half:

$$E_{i+1/2}\,\phi_{i+1/2}^{LO} - E_i\,\phi_i^{LO} + \frac{1}{2}\Sigma_{t,i}\,\Delta x_i\,J_{i+1/2}^{LO} = 0 \tag{23}$$

Boundary conditions in discrete space correspondingly take the form:

$$J_{1/2}^{LO} = C_L\big(\phi_{1/2}^{LO} - \phi_{in}^{+}\big) + J_{in}^{+} \tag{24}$$

$$J_{N+1/2}^{LO} = C_R\big(\phi_{N+1/2}^{LO} - \phi_{in}^{-}\big) + J_{in}^{-} \tag{25}$$

Instead of sweeping through the mesh like how the HO solution is solved, the LO equations can be cast into a linear system and solved with any standard linear solver. To do so, we rearrange them into a set of equations that defines the coupling of the cell and edge scalar flux unknowns only. First, the first spatial moment equations are rearranged to isolate the edge current:

$$J_{i-1/2}^{LO} = -\frac{2}{\Sigma_{t,i}\,\Delta x_i}\left(E_i\,\phi_i^{LO} - E_{i-1/2}\,\phi_{i-1/2}^{LO}\right) \tag{26}$$

$$J_{i+1/2}^{LO} = \frac{2}{\Sigma_{t,i}\,\Delta x_i}\left(E_i\,\phi_i^{LO} - E_{i+1/2}\,\phi_{i+1/2}^{LO}\right) \tag{27}$$

After substituting the new forms of the edge currents into the balance equation (Equation 21) and combining like terms, we arrive at the balance equation in terms of ϕ_i^{LO} and $\phi_{i\pm1/2}^{LO}$:

$$-\frac{2}{\Sigma_{t,i}\,\Delta x_i}\,E_{i+1/2}\,\phi_{i+1/2}^{LO} + \left(\frac{4}{\Sigma_{t,i}\Delta x_i}E_i + \Sigma_{r,i}\Delta x_i\right)\phi_i^{LO} - \frac{2}{\Sigma_{t,i}\Delta x_i}E_{i-1/2}\,\phi_{i-1/2}^{LO} = q_i\,\Delta x_i \tag{28}$$

$$\Sigma_{r,i} = \Sigma_{t,i} - \Sigma_{s,i} \tag{29}$$

In this scheme, we specify that the current at a cell interface is continuous, so setting $J_{i+1/2}^{LO} = J_{(i+1)-1/2}^{LO}$ leads to the current continuity condition:

$$\frac{2}{\Sigma_{t,i}\Delta x_i}\left(E_i\phi_i^{LO} - E_{i+1/2}\,\phi_{i+1/2}^{LO}\right) = \frac{2}{\Sigma_{t,i+1}\Delta x_{i+1}}\left(E_{i+1/2}\phi_{i+1/2}^{LO} - E_{i+1}\,\phi_{i+1}^{LO}\right) \tag{30}$$

Cleaning up Equation 30 leads to the second equation that defines neighbor cell coupling:

$$\frac{2}{\Sigma_{t,i+1}\,\Delta x_{i+1}}\,E_{i+1}\,\phi_{i+1}^{LO} + \frac{2}{\Sigma_{t,i}\,\Delta x_i}E_i\,\phi_i^{LO} - \left(\frac{2}{\Sigma_{t,i}\Delta x_i} + \frac{2}{\Sigma_{t,i+1}\,\Delta x_{i+1}}\right)E_{i+1/2}\,\phi_{i+1/2}^{LO} = 0 \tag{31}$$

Then for the boundary cells, the half-cell current equation can be used with the corresponding boundary closure equation. For the left edge:

$$\frac{2}{\Sigma_{t,1}\,\Delta x_1}\left(E_{1/2}\,\phi_{1/2}^{LO} - E_1\,\phi_1^{LO}\right) = C_L\left(\phi_{1/2}^{LO} - \phi_{in}^{+}\right) + J_{in}^{+} \tag{32}$$

Moving the unknowns to the LHS:

$$\left(\frac{2}{\Sigma_{t,1}\,\Delta x_1}E_{1/2} - C_L\right)\phi_{1/2}^{LO} - \frac{2}{\Sigma_{t,1}\,\Delta x_1}E_1\phi_1^{LO} = J_{in}^{+} - C_L\,\phi_{in}^{+} \tag{33}$$

where

$$J_{in}^{+} = \sum_{\mu_m>0} w_m\,\mu_m\,\psi_m^{in} \qquad , \qquad \phi_{in}^{+} = \sum_{\mu_m>0} w_m\psi_m^{in} \tag{34}$$

Repeating the process for the right edge, we obtain the following boundary equation:

$$\frac{2}{\Sigma_{t,N}\,\Delta x_N}E_N\phi_N^{LO} - \left(\frac{2}{\Sigma_{t,N}\,\Delta x_N}E_{N+1/2} + C_R\right)\phi_{N+1/2}^{LO} = J_{in}^{-} - C_R\,\phi_{in}^{-} \tag{35}$$

where

$$J_{in}^{-} = \sum_{\mu_m<0} w_m\,\mu_m\,\psi_m^{in} \qquad , \qquad \phi_{in}^{-} = \sum_{\mu_m<0} w_m\psi_m^{in} \tag{36}$$

For 1D, a tridiagonal matrix can be formed from Equation 28, Equation 31, Equation 33, and Equation 35 to solve for ϕ_i and $\phi_{i\pm1/2}$ simultaneously. Since the system is tridiaongal, the Thomas Algorithm can be employed solve the matrix extremely quickly. After solving for the cell and edge scalar fluxes, the edge currents can be easily reconstructed from Equation 26 and Equation 27.

2.3 Mapping of Low-order QD Grid Functions onto High-order Transport Grid Functions

After solving the LOQD equations, the LO solution must be mapped back onto the transport solution. Convergence performance can be sensitive to how the LO solution is prolonged onto the transport solution. In the case of the LLD equations, a prolongation operator must be defined for both the balance and first Legendre moment to have effective convergence. The chosen prolongation is:

$$\phi_i^{(l)} = \phi_i^{LO} \tag{37}$$

$$\widehat{\phi}_i^{(l)} = \frac{1}{2}\left(\phi_{i+1/2}^{LO} - \phi_{i-1/2}^{LO}\right) \tag{38}$$

For the zeroth moment, the transport cell average flux simply takes the cell average from the LOQD solution. The first moment is taken from the linear profile through the LOQD edge fluxes (Eq. (77) of [1]). A full derivation for the mapping of the LOQD solution onto the HO scattering be found in [1].

2.4 Solution Algorithm

From Equation 3, it is clear that the loss and source are a function of the angular flux solution. Therefore, when numerically solving the transport equation, the source must be iterated upon. The most straightforward method is simply the source iteration method (SI), in which the RHS of of Equation 3 is lagged until the flux solution converges (to within a specified tolerance). For the HOLO scheme using QD, the LO (QD) solve is performed following a HO mesh sweep for each source iteration. The HOLO algorithm is provided in algorithm 1.

Algorithm 1 Source (inner) Iteration Method with Quasi-Diffusion

Guess initial angular and scalar flux
Set initial QD factors: $E_i = E_{i+1/2} = 1/3$, $C_L = -1/2$, $C_R = 1/2$
while $l \leq l_{max}$ **do**
  $l = l+1$
  Compute new source for each cell
  for i = 1 to N_{cells} **do**
    Compute source term $Q_i^{(l)}$ and $\widehat{Q}_i^{(l)}$ from $\phi_i^{(l-1)}$, $\widehat{\phi}_i^{(l-1)}$
  end for
  Sweep in every ordinate direction
  for μ_m in quadrature angles **do**
    if $\mu_m > 0$ **then**
      Start sweep from x=0
      $\psi(0, \mu_m)^{(l)} = \psi_{in}^{+}(\mu_m)$
      for i = 1 to N_{cells} **do**
        Solve for $\psi_{m,i}^{(l)}$, $\widehat{\psi}_{m,i}^{(l)}$, and $\psi_{m,i+1/2}^{(l)}$ using Equation 9
      end for
    else if $\mu_m < 0$ **then**
      Start sweep from x=X
      $\psi(X, \mu_m)^{(l)} = \psi_{in}^{-}(\mu_m)$
      for i = N_{cells} to 1 **do**
        Solve for $\psi_{m,i}^{(l)}$, $\widehat{\psi}_{m,i}^{(l)}$, and $\psi_{m,i-1/2}^{(l)}$ using Equation 13
      end for
    end if
  end for
  Perform QD acceleration to obtain LO scalar flux
  if $l > 1$ **then**
    Compute Eddington factors from the high-order solution (E)
    Compute boundary factors from the outgoing angular flux (C_L, C_R)
  end if
  Compute incoming partial moments $\phi_{in}^{\pm}$, $J_{in}^{\pm}$ from the boundary conditions
  Assemble the tridiagonal LOQD matrix
  Solve the system with the Thomas algorithm to obtain ϕ_i^{LO} and $\phi_{i+1/2}^{LO}$
  Prolong low order data to high-order system:
  for i = 1 to N_{cells} **do**
    $\phi_i^{(l)} = \phi_i^{LO}$
    $\widehat{\phi}_i^{(l)} = \frac{1}{2}\left(\phi_{i+1/2}^{LO} - \phi_{i-1/2}^{LO}\right)$
  end for
  if $l > 2$ **then**
    Compute spectral radius of iteration: $\rho_\phi^{(l)} = \frac{||\phi^{(l)} - \phi^{(l-1)}||_\infty}{||\phi^{(l-1)} - \phi^{(l-2)}||_\infty}$
  else
    $\rho_\phi^{(l)} = 1/2$
  end if
  if $||\phi^{(l)} - \phi^{(l-1)}||_\infty < \epsilon_\phi\left(\frac{1}{\rho_\phi^{(l)}} - 1\right)$ **then**
    Solution converged.
  end if
end while

3 Results

3.1 Test A

Test is A is a heterogeneous fixed source problem with non-multiplying medium. Two total material regions exist. An incoming angular flux boundary conditions are prescribed on the left side with of the domain vacuum boundary on the right.

$$\psi(0, \mu) = 1.0 \quad \mu > 0$$

A non-uniform spatial mesh is used with 10 cells in the first region with width 0.10 cm and 10 cells in the second region with width 1.0 cm. The Double S_4 Gauss-Legendre quadrature set was chosen, and a convergence criterion on the flux solution of $\epsilon_\phi = 10^{-10}$. The material regions and cell edge locations are illustrated in Figure 1. The material cross sections and subdomain regions are provided in Table 1.

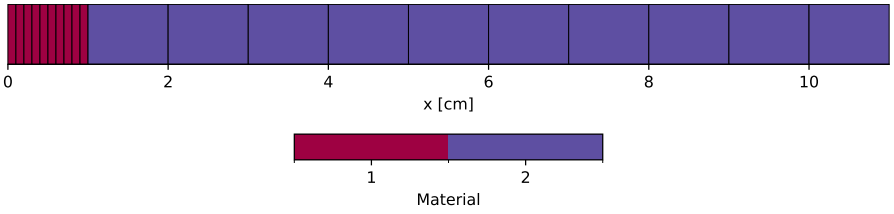


Figure 1: Test A mesh

Table 1: Test A material region definition

Material	Right edge [cm]	Σ_t [cm^{-}1]	Σ_s [cm^{-1}]	Σ_f [cm^{-1}]	ν [neutrons per fission]	q [neutrons/cm^3·sec]
1	1.0	2.0	0.0	0.0	0.0	0.0
2	11.0	100.0	100.0	0.0	0.0	0.0

After solving the system with the with LLD discretization and QD acceleration, the solution converged in $\boxed{\text{N}_{iter} = 7 \text{ iterations}}$. This is a significant improvement over the unaccelerated case using LD discretization. The speedup is a result of QD acceleration providing a very good approximation of the scalar flux in the highly diffusive region. The values for the converged boundary factors are $\boxed{\text{C}_L = \text{-0.80031 and C}_R = 0.57283}$. The spatial distributions of various high-order and low-order solution variables upon convergence are given in Figure 2 through Figure 5. The infinity norm of the cell average scalar flux versus source iteration is shown in Figure 10. The spectral radius of convergence with respect to source iteration is provided in Figure 11. A spectral radius close to unity implies that a solution is difficult to converge. As opposed to the unaccelerated case, the spectral radius of the problem quickly approaches 0.1 highlighting the observed fast convergence.

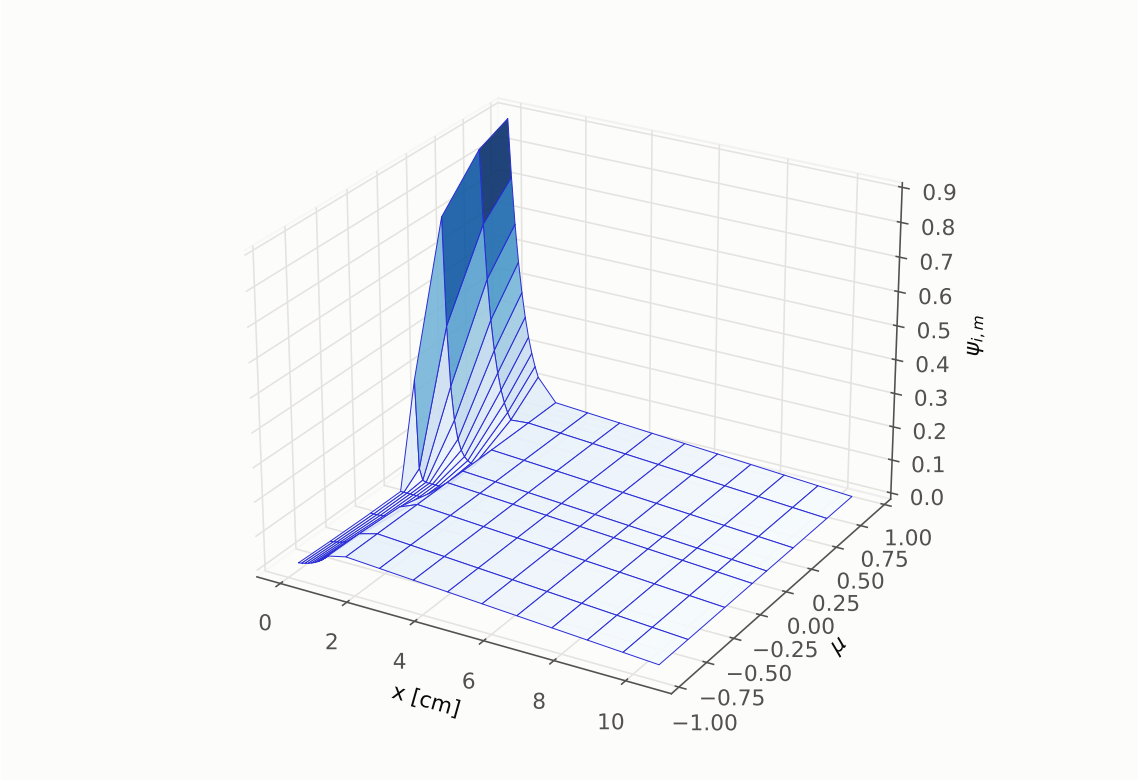


Figure 2: Test A cell average angular flux versus position and angle (x_i, μ_m)

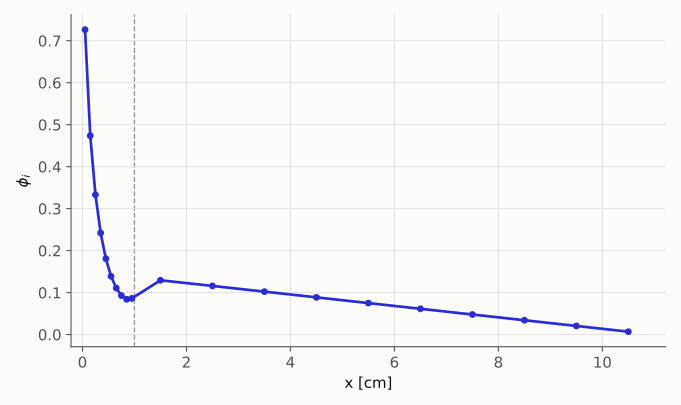


Figure 3: Test A cell average scalar flux from the low-order QD solution

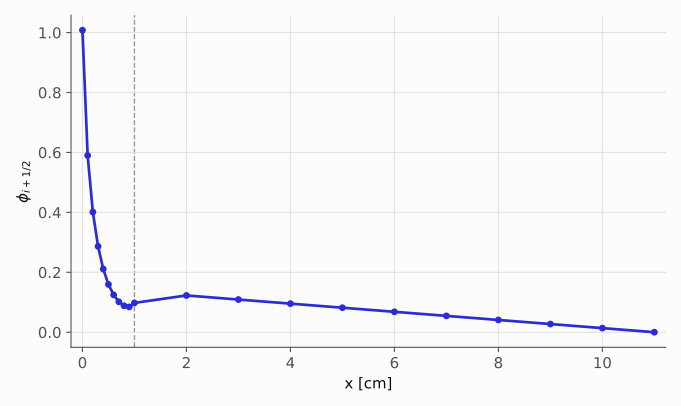


Figure 4: Test A cell edge scalar flux from the low-order QD solution

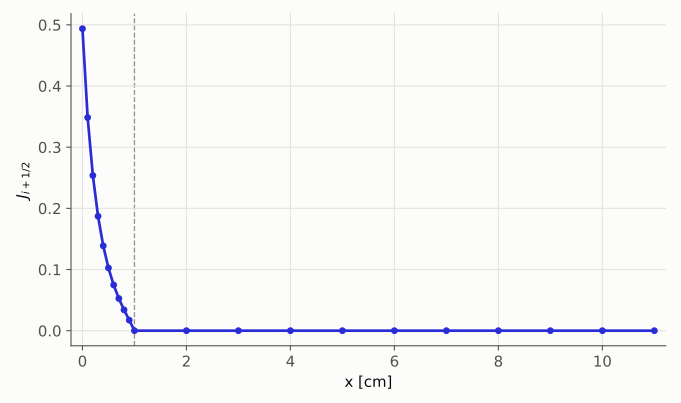


Figure 5: Test A cell edge current from the low-order QD solution

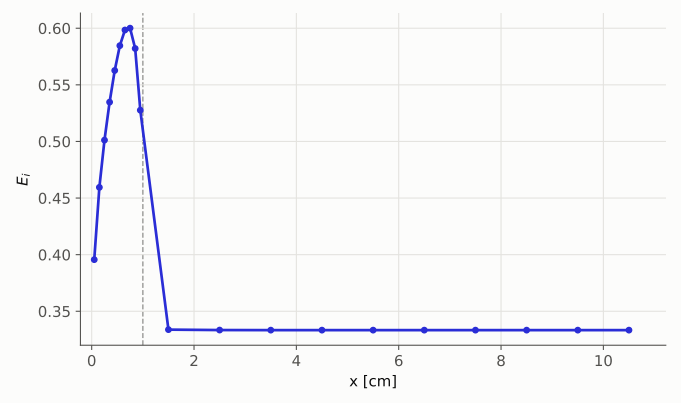


Figure 6: Test A cell average Eddington factor from the low-order QD solution

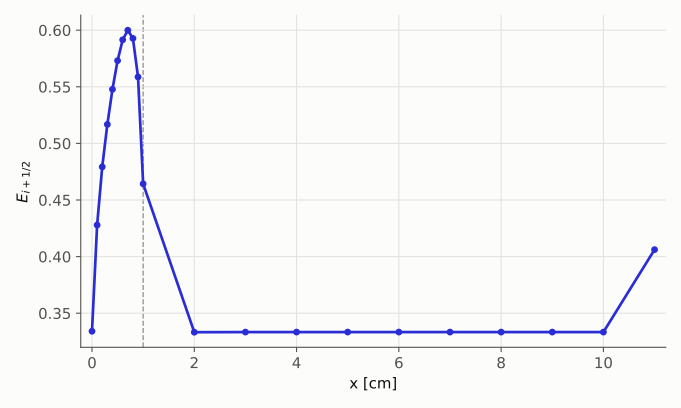


Figure 7: Test A cell edge Eddington factor from the low-order QD solution

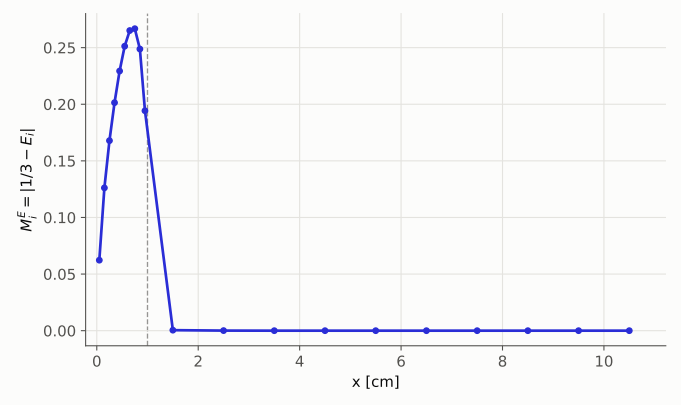


Figure 8: Test A cell average metric (M^E) of transport effects

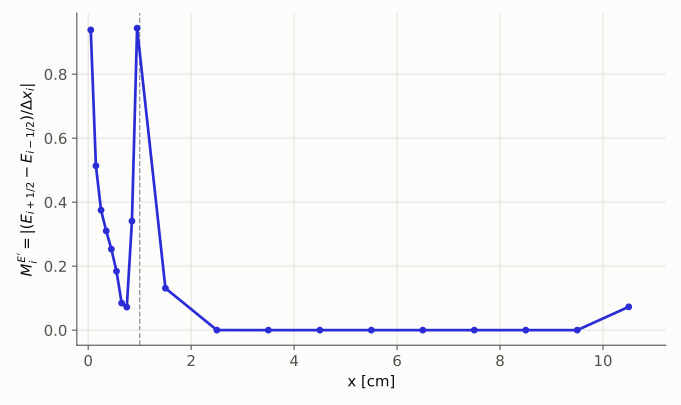


Figure 9: Test A cell average metric ($M^{E'}$) of transport effects

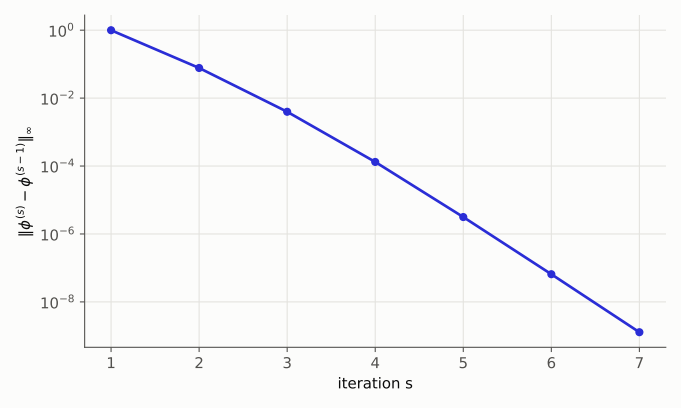


Figure 10: Test A convergence history versus iteration

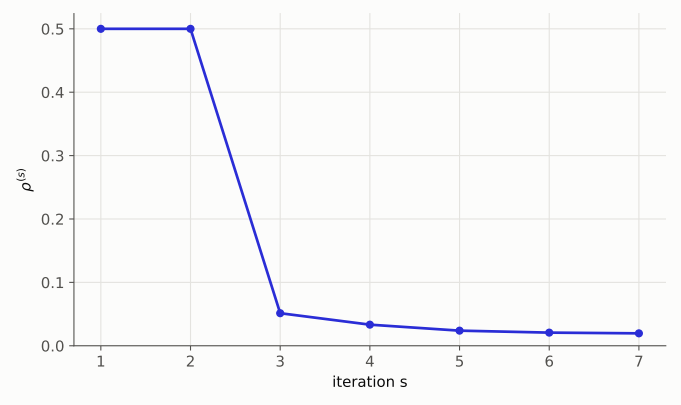


Figure 11: Test A spectral radius versus iteration

3.2 Test B

Test is B is a heterogeneous fixed source problem with non-multiplying medium. Two total material regions exist. Vacuum boundary conditions are prescribed on both edges of the slab. A non-uniform spatial mesh is used with 10 cells in the first region with width 1.0 cm and 40 cells in the second region with width 0.5 cm. The Double S_4 Gauss-Legendre quadrature set was chosen, and a convergence criterion on the flux solution of $\epsilon_\phi = 10^{-10}$. The material regions and cell edge locations are illustrated in Figure 12. The material cross sections and subdomain regions are provided in Table 2.

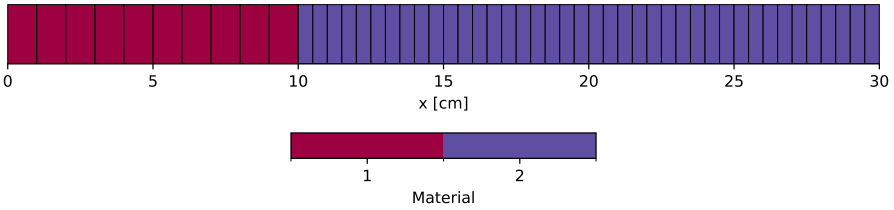


Figure 12: Test B mesh

Table 2: Test B material region definition

Material	Right edge [cm]	Σ_t [cm^{-1}]	Σ_s [cm^{-1}]	Σ_f [cm^{-1}]	ν [neutrons per fission]	q [neutrons/cm^3·sec]
1	10.0	1.0	0.5	0.0	0.0	0.5
2	30.0	10.0	9.9	0.0	0.0	0.01

Test B converged in $\boxed{\text{N}_{iter} = 9 \text{ iterations}}$. The values for the converged boundary factors are $\boxed{\text{C}_L = \text{-0.51248 and C}_R = 0.53197}$. The spatial distributions of various high-order and low-order solution variables upon convergence are given in Figure 13 through Figure 16. The infinity norm of the cell average scalar flux versus source iteration is shown in Figure 21. The spectral radius of convergence with respect to source iteration is provided in Figure 22.

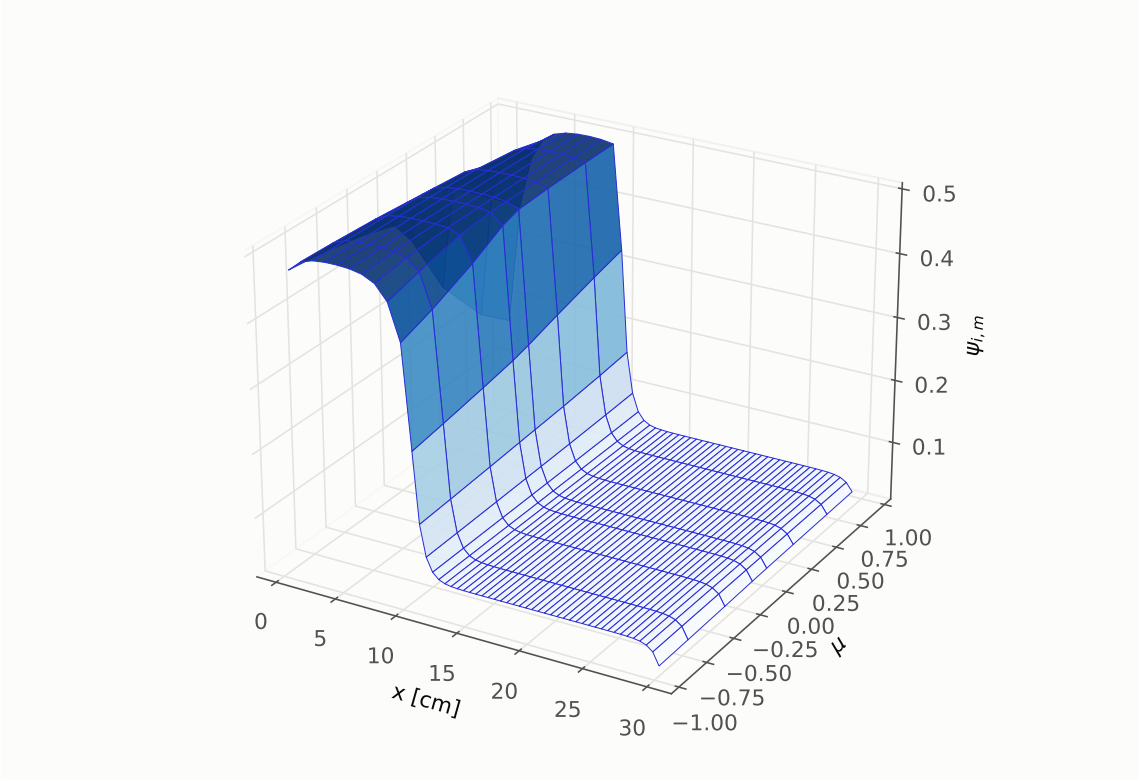


Figure 13: Test B cell average angular flux versus position and angle (x_i, μ_m)

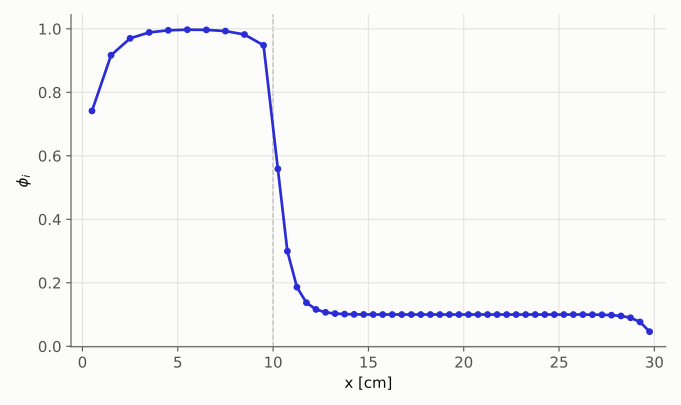


Figure 14: Test B cell average scalar flux from the low-order QD solution

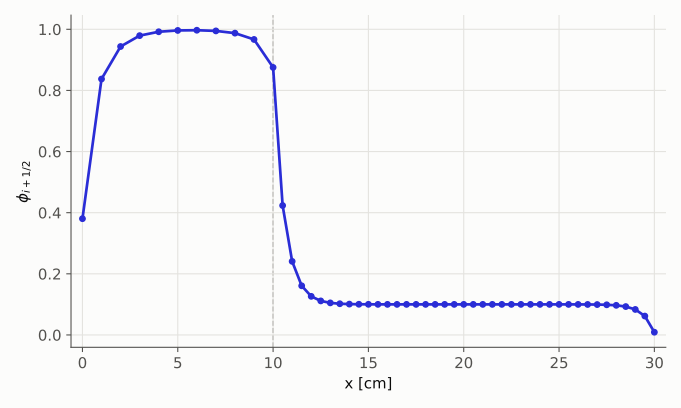


Figure 15: Test B cell edge scalar flux from the low-order QD solution

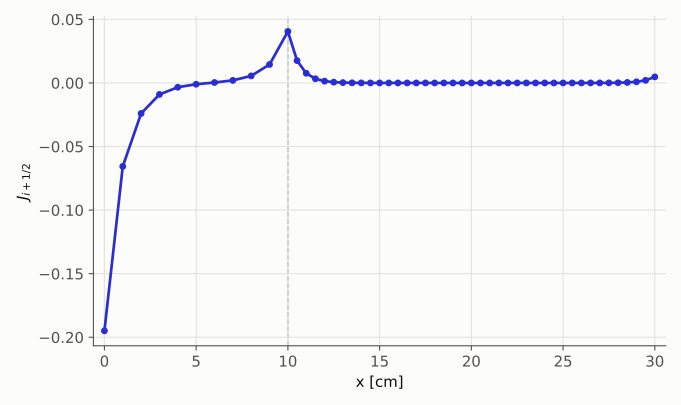


Figure 16: Test B cell edge current from the low-order QD solution

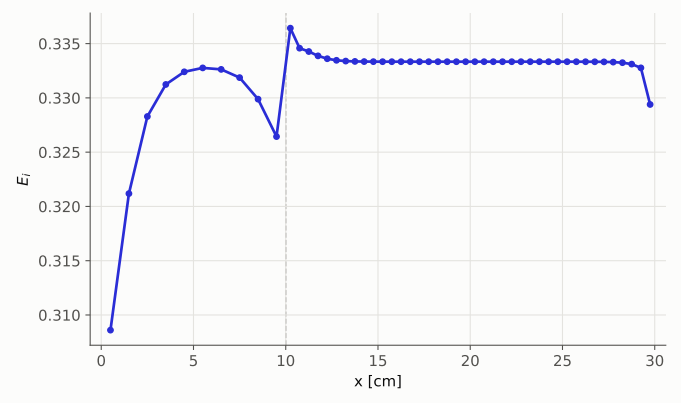


Figure 17: Test B cell average Eddington factor from the low-order QD solution

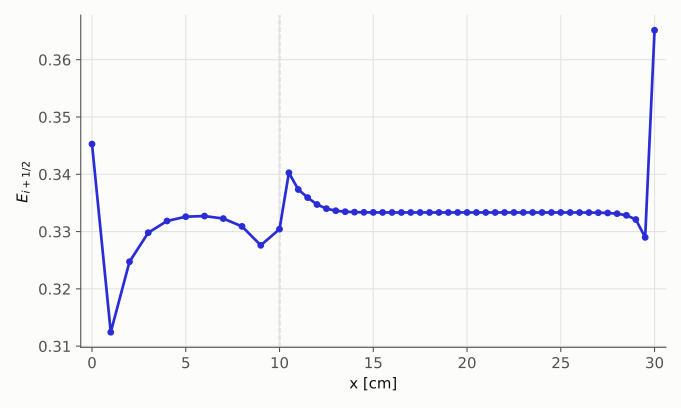


Figure 18: Test B cell edge Eddington factor from the low-order QD solution

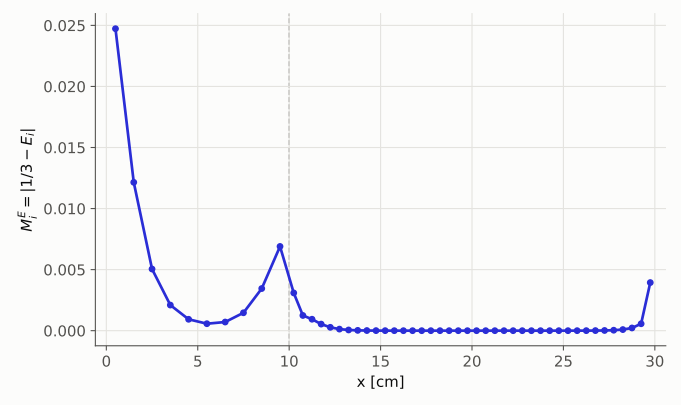


Figure 19: Test B cell average metric (M^E) of transport effects

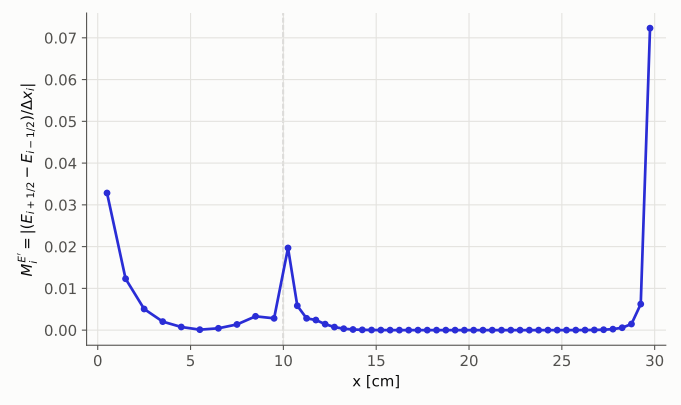


Figure 20: Test B cell average metric ($M^{E'}$) of transport effects

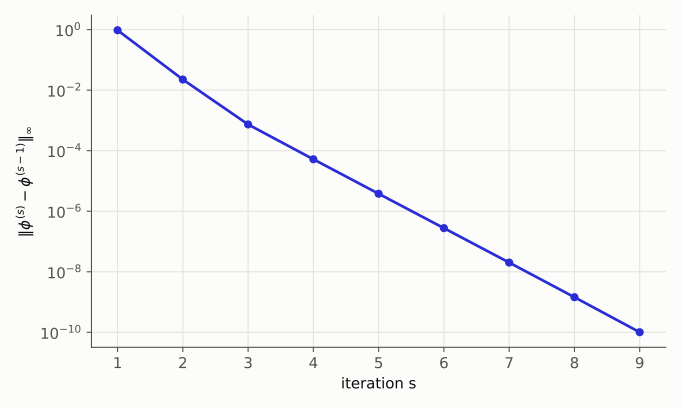


Figure 21: Test B convergence history versus iteration

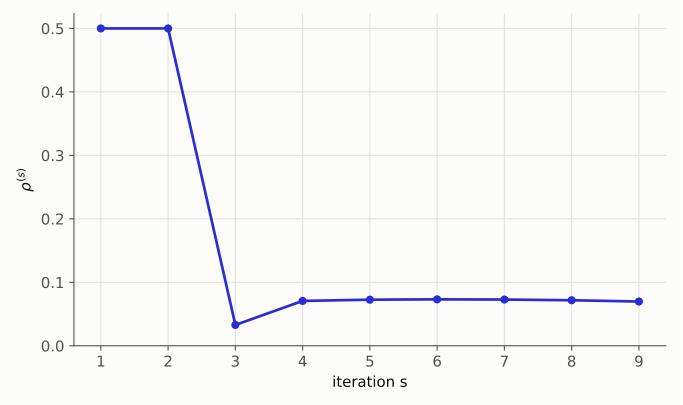


Figure 22: Test B spectral radius versus iteration

3.3 Test C

Test is C is a heterogeneous fixed source problem with non-multiplying medium. Two total material regions exist. Vacuum boundary conditions are prescribed on both edges of the slab. A non-uniform spatial mesh is used with 10 cells in the first region with width 1.0 cm and 40 cells in the second region with width 0.5 cm. The Double S_4 Gauss-Legendre quadrature set was chosen, and a convergence criterion on the flux solution of $\epsilon_\phi = 10^{-10}$. The material regions and cell edge locations are illustrated in Figure 23. The material cross sections and subdomain regions are provided in Table 3.

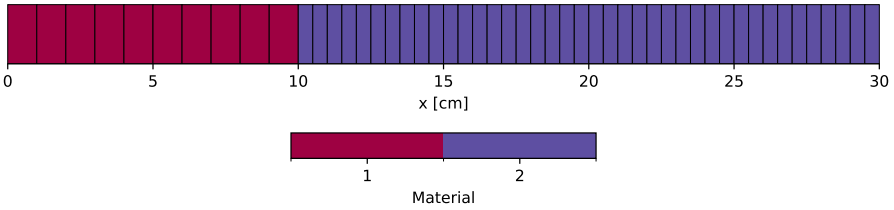


Figure 23: Test C mesh

Table 3: Test C material region definition

Material	Right edge [cm]	Σ_t [cm^{-1}]	Σ_s [cm^{-1}]	Σ_f [cm^{-1}]	ν [neutrons per fission]	q [neutrons/cm^3·sec]
1	10.0	1.0	0.5	0.0	0.0	0.5
2	30.0	10.0	9.9	0.0	0.0	0.0

Test C converged in $\boxed{N_{iter} = 8 \text{ iterations}}$. The values for the converged boundary factors are $\boxed{C_L = -0.51248 \text{ and } C_R = 0.64831}$. The spatial distributions of various high-order and low-order solution variables upon convergence are given in Figure 24 through Figure 27. The infinity norm of the cell average scalar flux versus source iteration is shown in Figure 32. The spectral radius of convergence with respect to source iteration is provided in Figure 33.

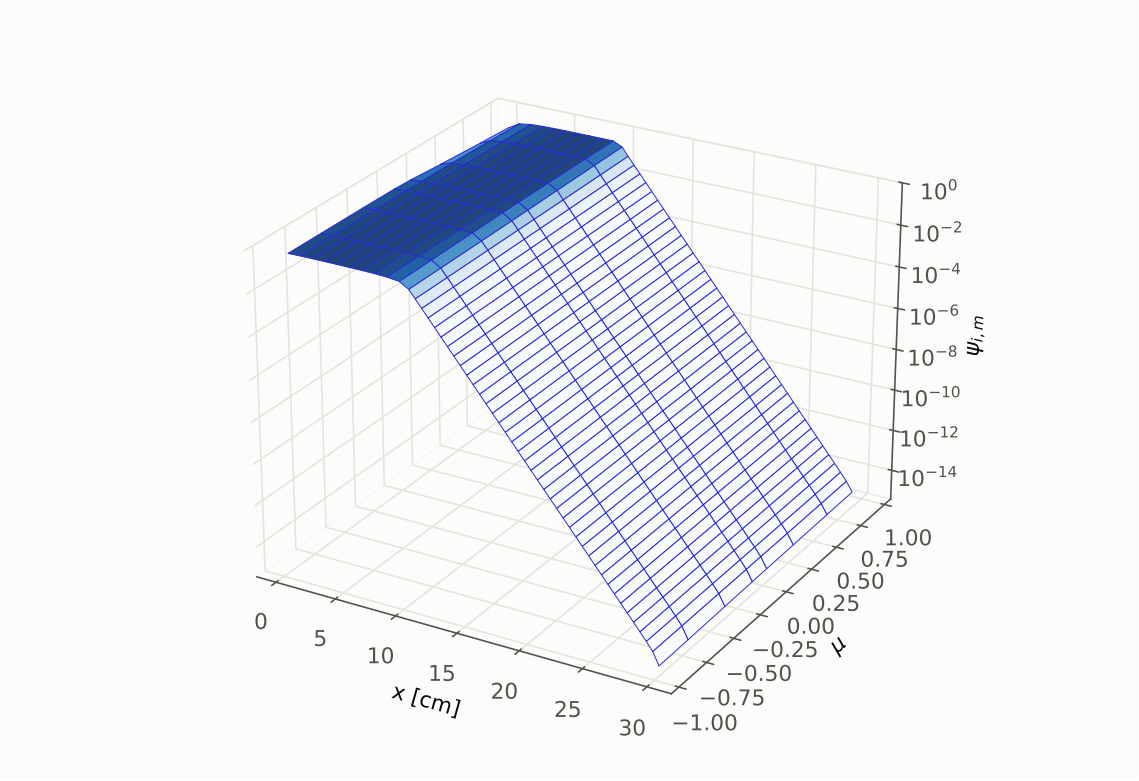


Figure 24: Test C cell average angular flux versus position and angle (x_i, μ_m)

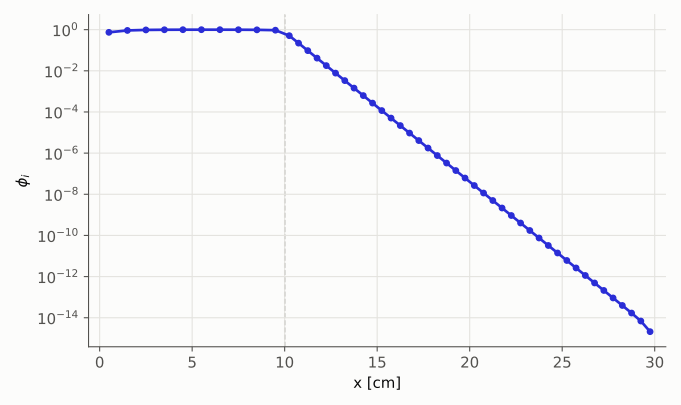


Figure 25: Test C cell average scalar flux from the low-order QD solution

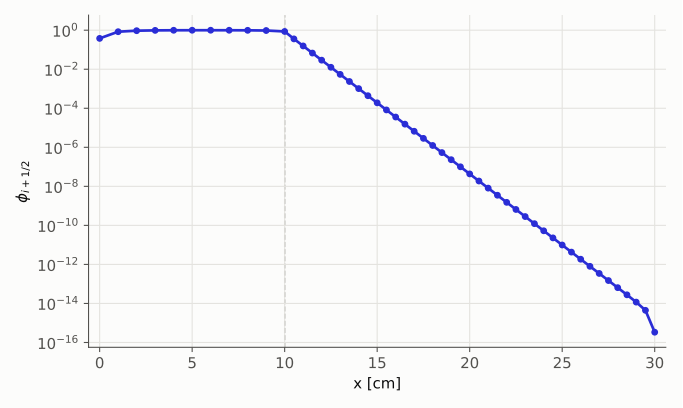


Figure 26: Test C cell edge scalar flux from the low-order QD solution

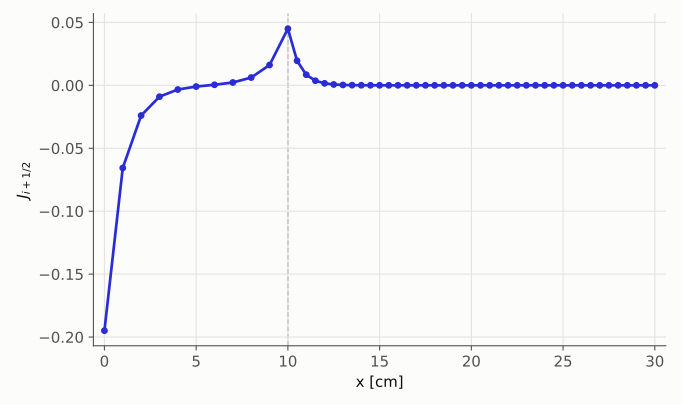


Figure 27: Test C cell edge current from the low-order QD solution

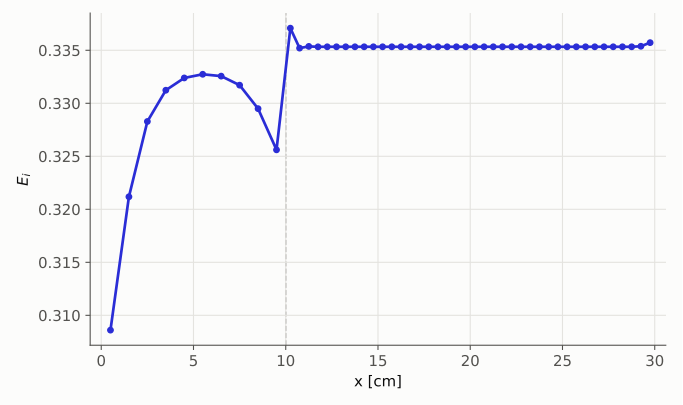


Figure 28: Test C cell average Eddington factor from the low-order QD solution

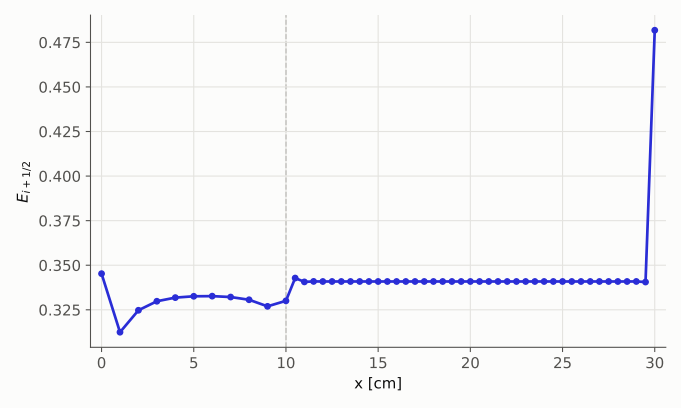


Figure 29: Test C cell edge Eddington factor from the low-order QD solution

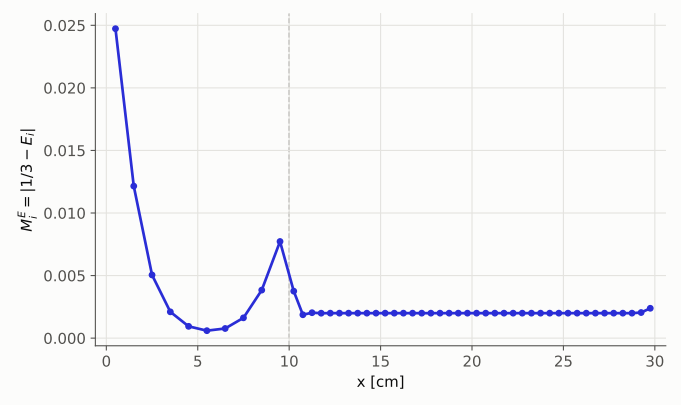


Figure 30: Test C cell average metric (M^E) of transport effects

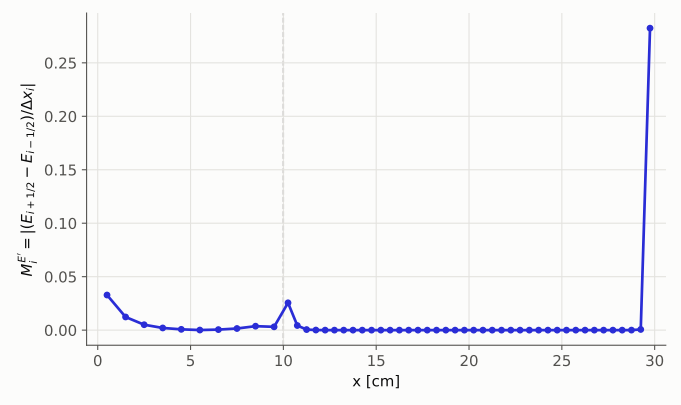


Figure 31: Test C cell average metric ($M^{E'}$) of transport effects

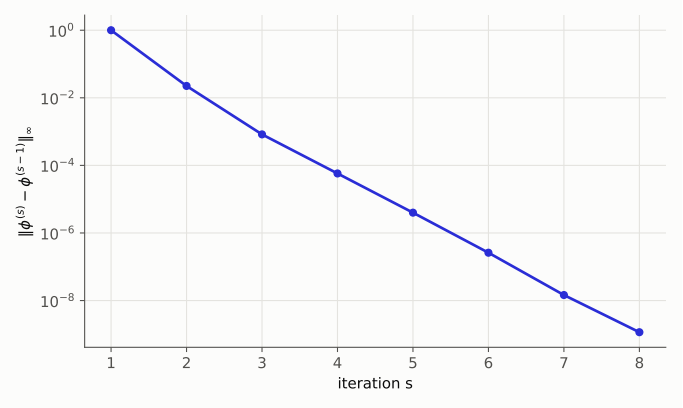


Figure 32: Test C convergence history versus iteration

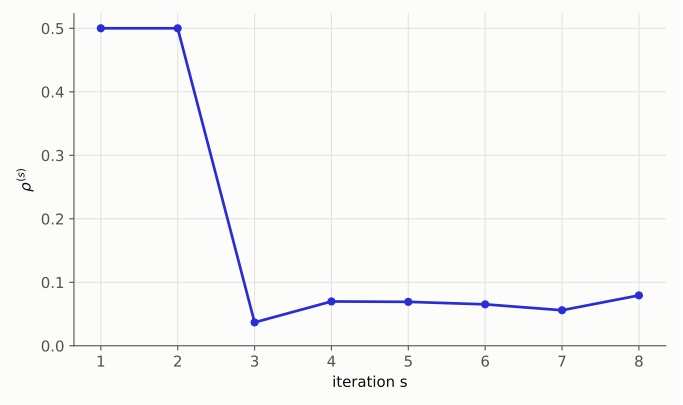


Figure 33: Test C spectral radius versus iteration

3.4 Test D

Test is D is a homogeneous fixed source problem with non-multiplying medium. Vacuum boundary conditions are prescribed on both edges of the slab. A uniform spatial mesh is used with 10 cells with width 0.1 cm. The Double S_4 Gauss-Legendre quadrature set was chosen, and a convergence criterion on the flux solution of $\epsilon_\phi = 10^{-10}$. For Test D, two configurations are investigated based on modified scattering. The material regions and cell edge locations are illustrated in Figure 23. The material cross sections and subdomain regions are provided in Table 4.

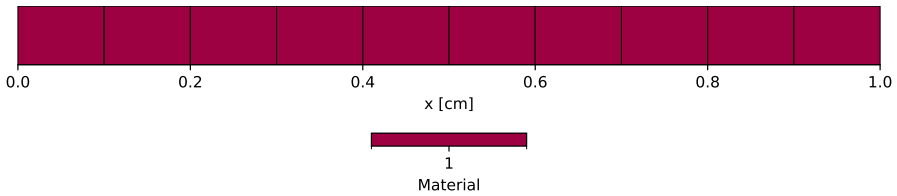


Figure 34: Test D mesh

Table 4: Test D material region definition, with $\Sigma_t = 1/\varepsilon$, $\Sigma_a = \varepsilon$ and $q = \varepsilon$ for (a) $\varepsilon = 10^{-2}$ and (b) $\varepsilon = 10^{-4}$ (the parameter is called ν in the problem statement)

Material	Right edge [cm]	Σ_t [cm^{-1}]	Σ_s [cm^{-1}]	Σ_f [cm^{-1}]	ν [neutrons per fission]	q [neutrons/cm^3·sec]
1 (a)	1.0	10^2	99.99	0.0	0.0	10^{-2}
1 (b)	1.0	10^4	9999.9999	0.0	0.0	10^{-4}

3.4.1 (a)

Test D(a) converged in $\boxed{N_{iter} = 6 \text{ iterations}}$. The values for the converged boundary factors are $\boxed{C_L = -0.54395 \text{ and } C_R = 0.54395}$. The spatial distributions of various high-order and low-order solution variables upon convergence are given in Figure 35 through Figure 38. The infinity norm of the cell average scalar flux versus source iteration is shown in Figure 43. The spectral radius of convergence with respect to source iteration is provided in Figure 44.

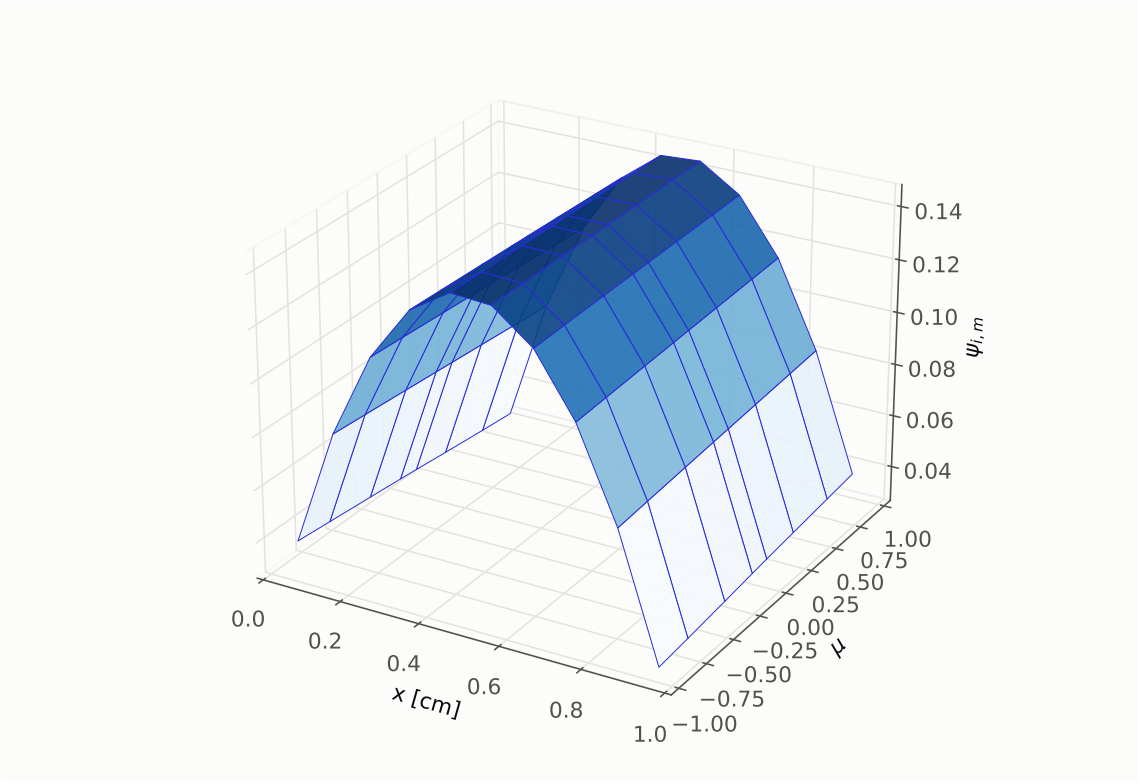


Figure 35: Test D(a) cell average angular flux versus position and angle (x_i, μ_m)

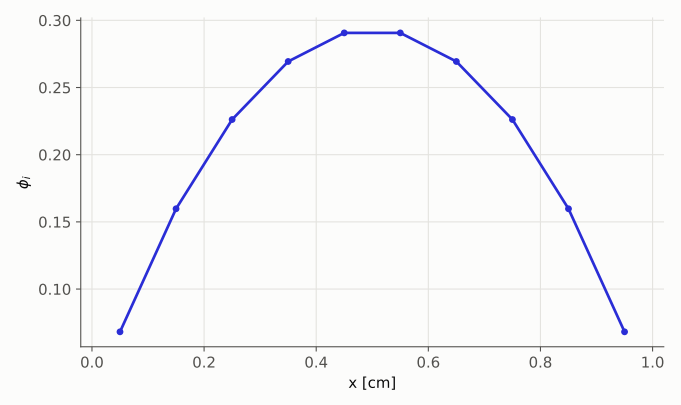


Figure 36: Test D(a) cell average scalar flux from the low-order QD solution

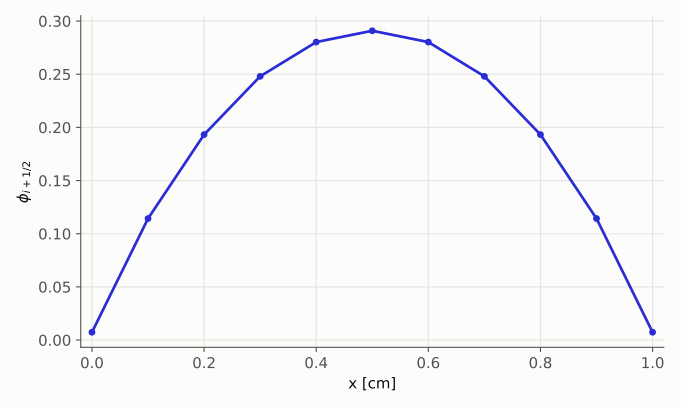


Figure 37: Test D(a) cell edge scalar flux from the low-order QD solution

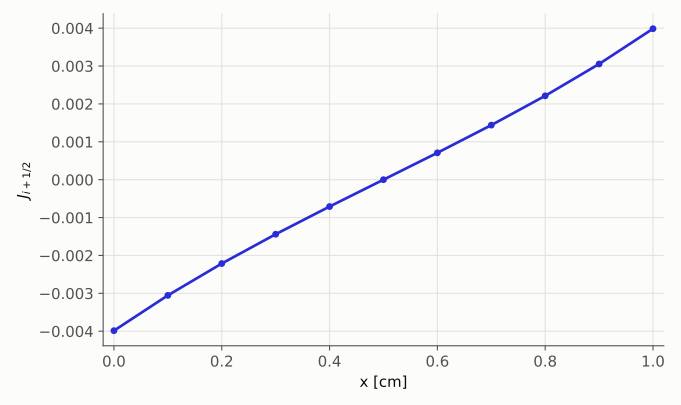


Figure 38: Test D(a) cell edge current from the low-order QD solution

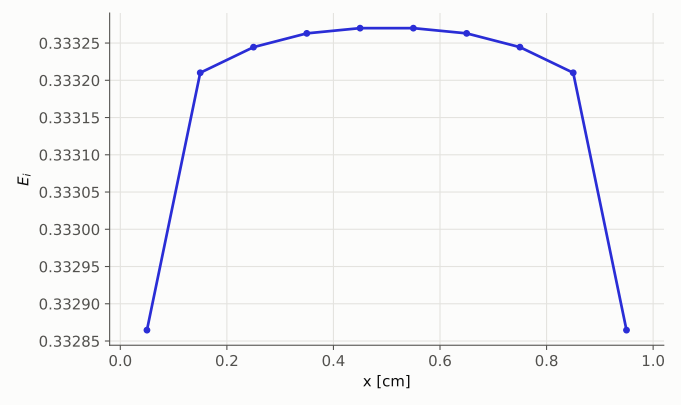


Figure 39: Test D(a) cell average Eddington factor from the low-order QD solution

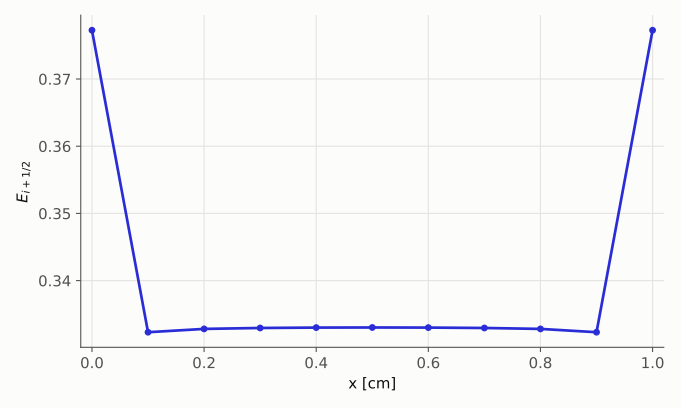


Figure 40: Test D(a) cell edge Eddington factor from the low-order QD solution

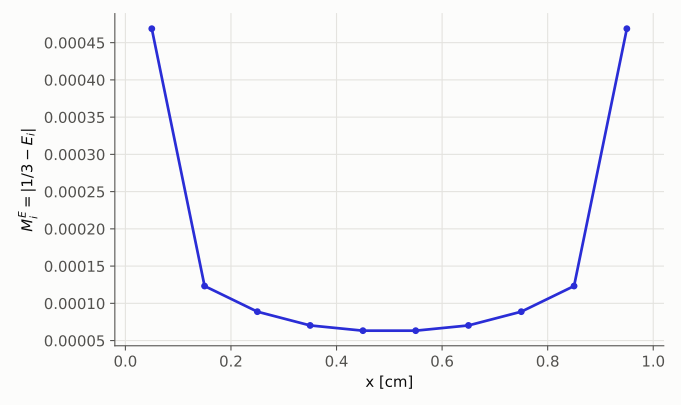


Figure 41: Test D(a) cell average metric (M^E) of transport effects

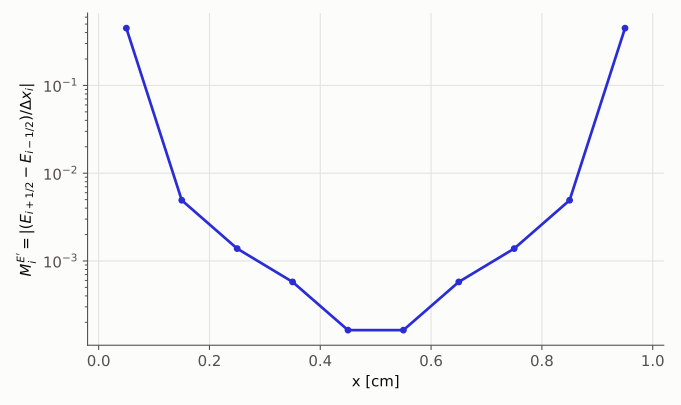


Figure 42: Test D(a) cell average metric ($M^{E'}$) of transport effects

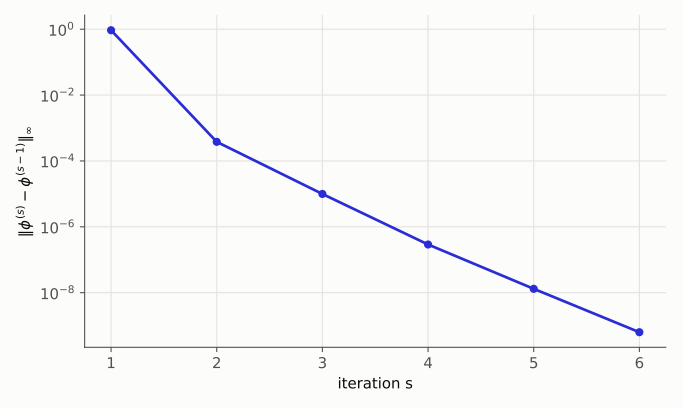


Figure 43: Test D(a) convergence history versus iteration

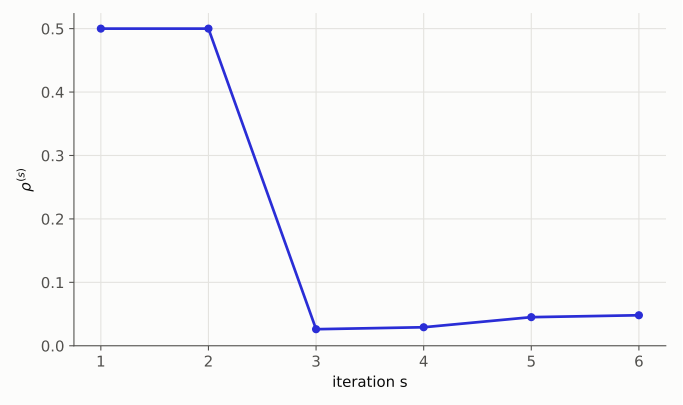


Figure 44: Test D(a) spectral radius versus iteration

3.4.2 (b)

Test D(b) in $\boxed{\text{N}_{iter} = 3 \text{ iterations}}$. The values for the converged boundary factors are $\boxed{\text{C}_L = \text{-0.50123 and C}_R = 0.50123}$. The spatial distributions of various high-order and low-order solution variables upon convergence are given in Figure 45 through Figure 48. The infinity norm of the cell average scalar flux versus source iteration is shown in Figure 53. The spectral radius of convergence with respect to source iteration is provided in Figure 54.

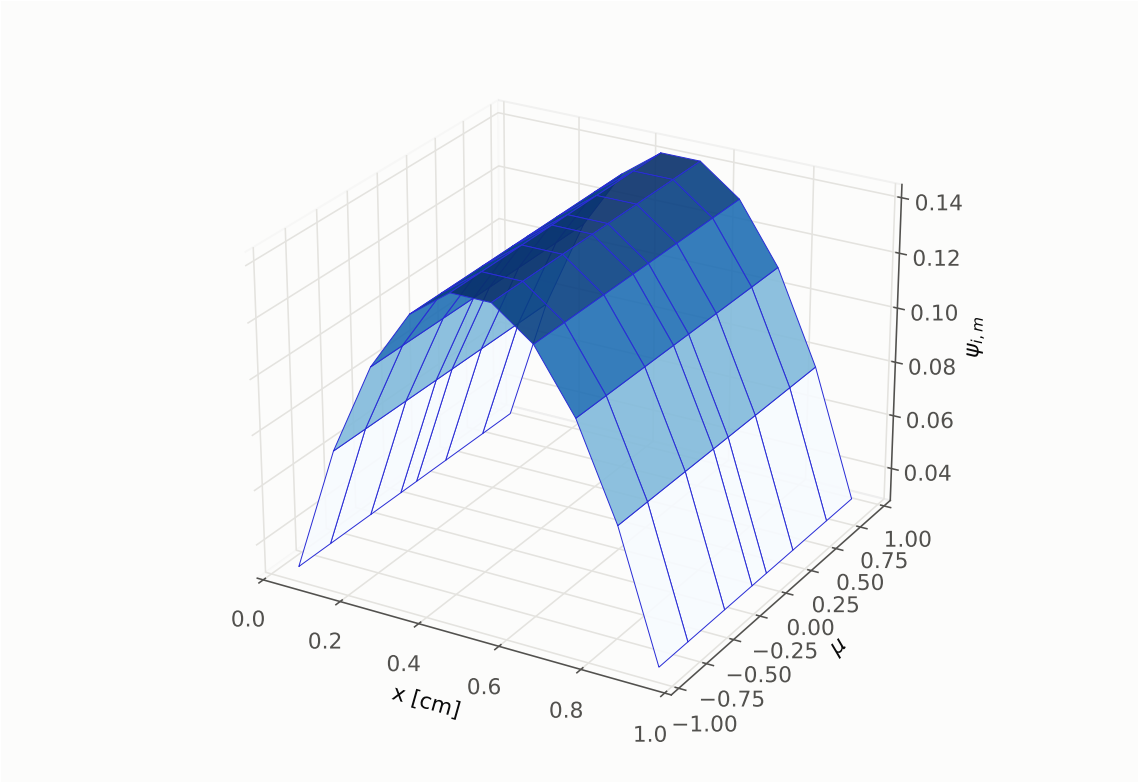


Figure 45: Test D(b) cell average angular flux versus position and angle (x_i, μ_m)

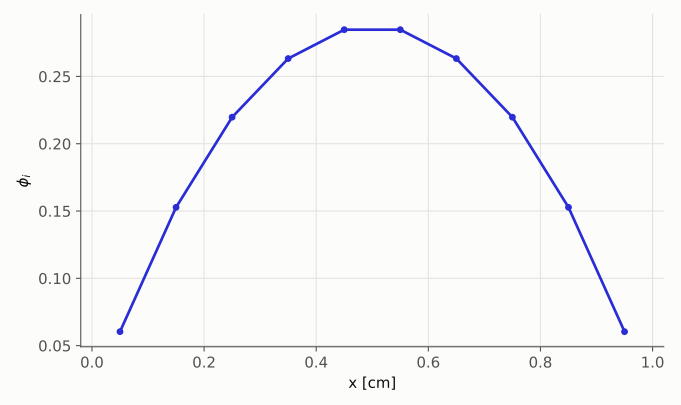


Figure 46: Test D(b) cell average scalar flux from the low-order QD solution

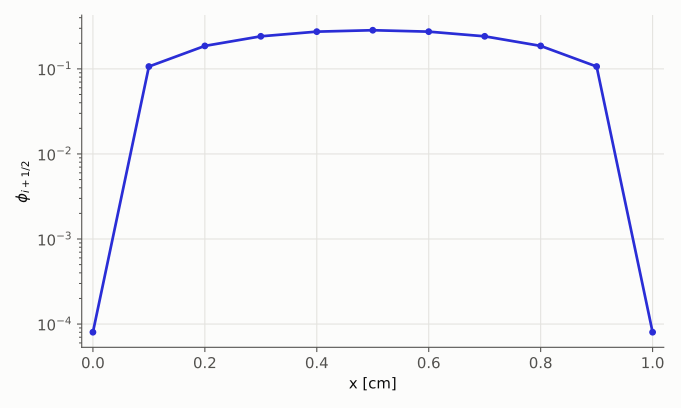


Figure 47: Test D(b) cell edge scalar flux from the low-order QD solution

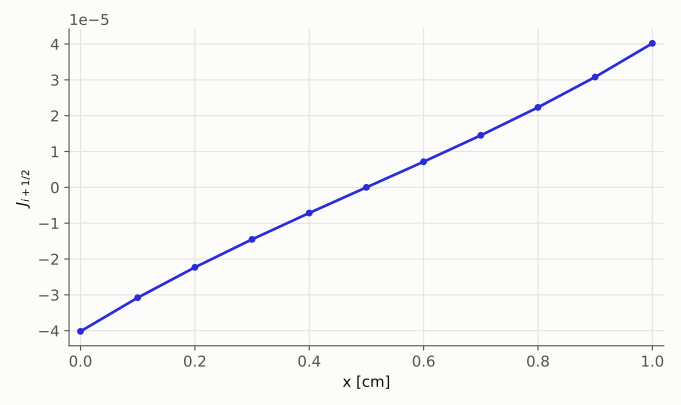


Figure 48: Test D(b) cell edge current from the low-order QD solution

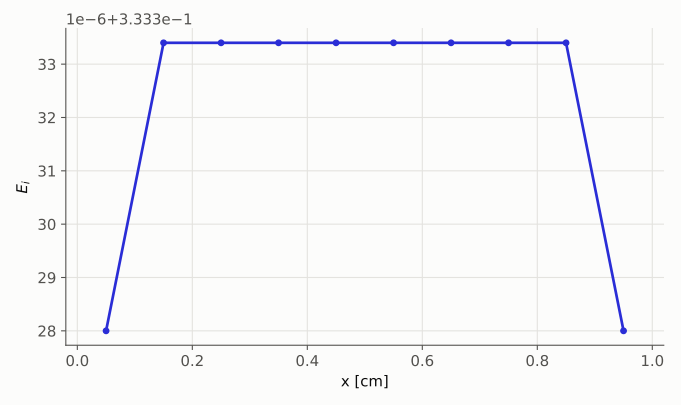


Figure 49: Test D(b) cell average Eddington factor from the low-order QD solution

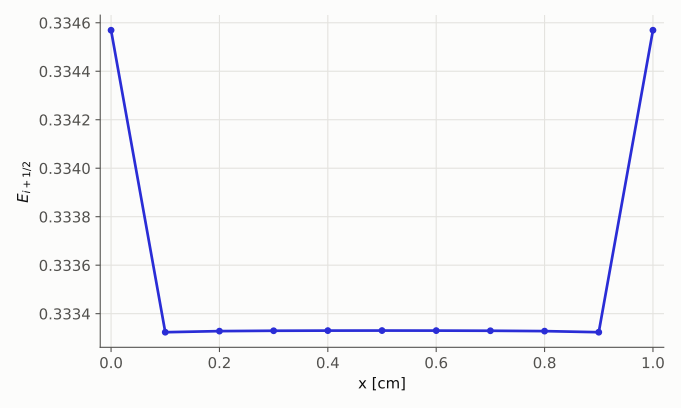


Figure 50: Test D(b) cell edge Eddington factor from the low-order QD solution

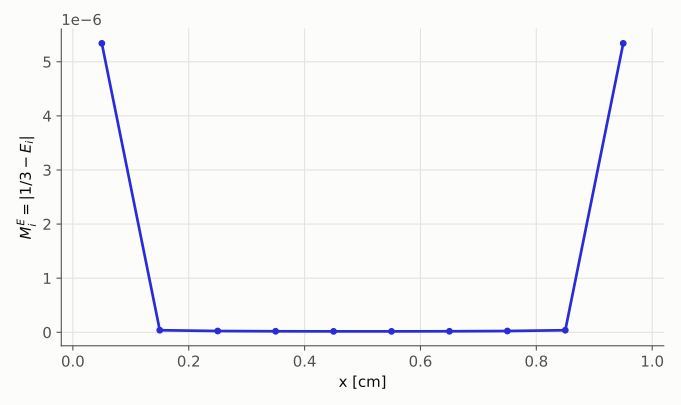


Figure 51: Test D(b) cell average metric (M^E) of transport effects

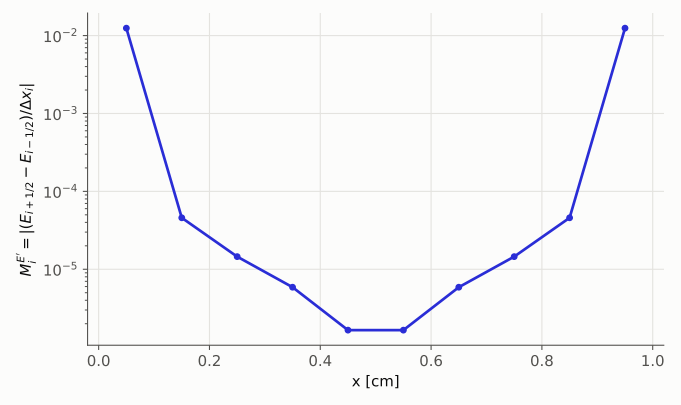


Figure 52: Test D(b) cell average metric ($M^{E'}$) of transport effects

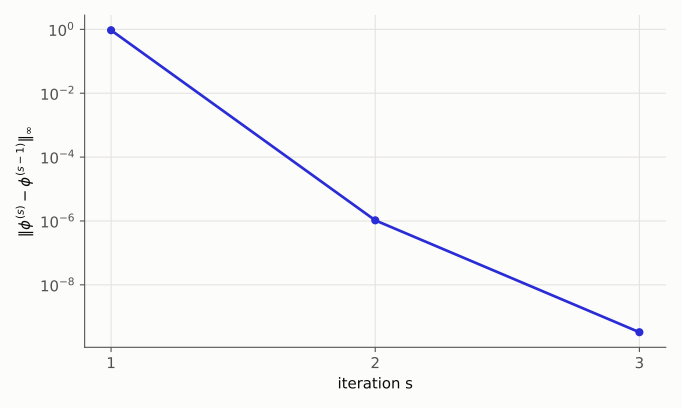


Figure 53: Test D(b) convergence history versus iteration

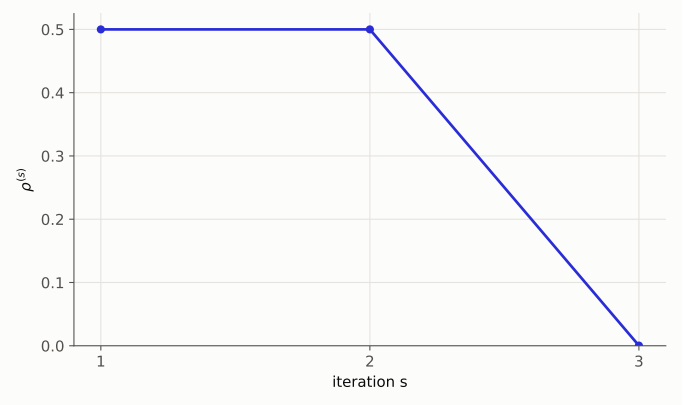


Figure 54: Test D(b) spectral radius versus iteration

4 Summary

The one-dimensional S_N transport solver developed in Project 1 was extended to using lumped linear discontinuous spatial discretization. Additionally, the Quasi-Diffusion Method was implemented to accelerate the convergence of the scattering source distribution. With QD, the transport solver now converges in a very small amount of source iterations, even in extremely high scattering and diffusive regions.

References

[1] J. S. WARSA and D. Y. ANISTRATOV, "Two-Level Transport Methods with Independent Discretization," Tech. Rep. LA-UR-18-22117, Los Alamos National Laboratory (2018).